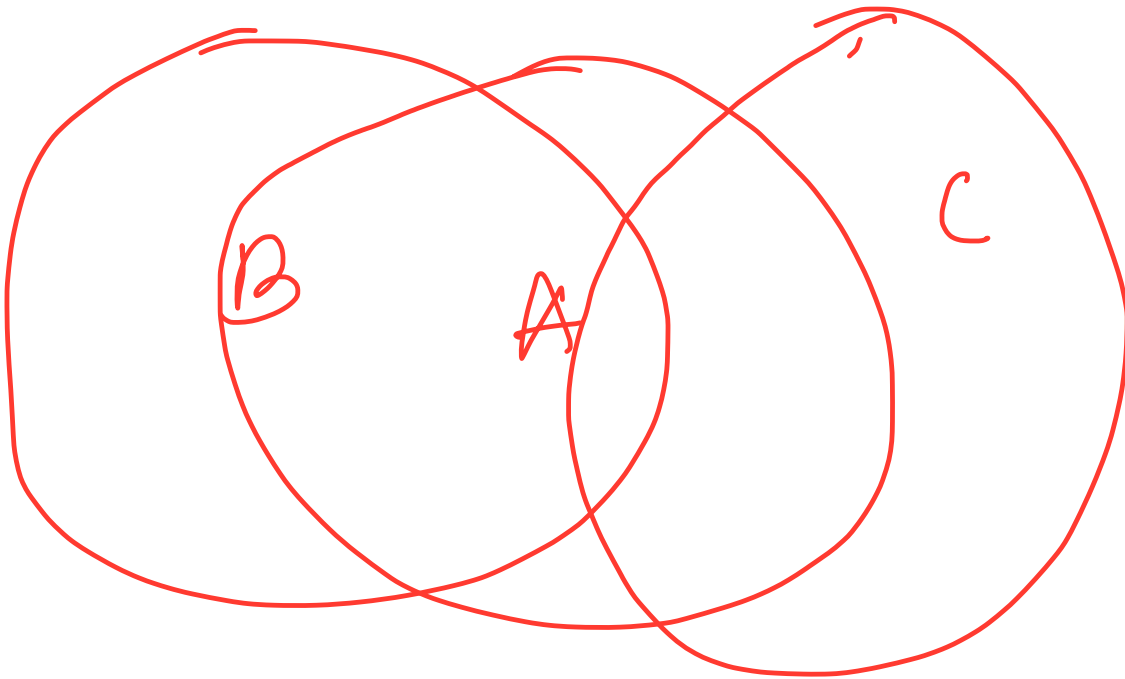


# Hidden Node Problem



RTS CTS

↓  
Request to Send

Clears to Send  
Before any data transmission

- Send a short control frame named RTS
- After successfully receiving RTS, the receiver sends CTS signal
- After getting CTS successfully, B transmits data packets
- , C defers its transmission

## Slow-talker problem

CSMA/CA  $\rightarrow$  provides equal # of opportunities in steady state to all nodes.

Now, if there is a node with bitrate poor, he will occupy the channel for more time  $\rightarrow$

10 # of opportu

A  $\rightarrow$  54 Mbps  $\rightarrow$  aintime occupied by B is  
B  $\rightarrow$  1 Mbps  $\rightarrow$  54 times longer than A

\* High density + WiFi does not scale well  
in a dense deployment

